3WD-VLA: Hierarchical Three-Way Decision for Collaborative Dual-View Robotic Grasping

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Abstract. Collaborative robotic manipulation in shared workspaces requires robots to decide not only how to grasp, but also whether to execute immediately, ask for clarification, plan a recovery action, or stop for safety. We present 3WD-VLA, a hierarchical dual-view framework that combines structured decision generation by a vision-language model with sequential three-way decision making. Given a top-view image, a wrist-view image, and a language instruction, the reasoning module outputs pairwise action preferences and safety reasoning. A supervisor first filters negative-domain cases such as hand intrusion and target absence. For the remaining scenes, the pairwise preferences between the direct action and the three alternative actions, namely ask, replan, and reject, are converted into latent action merits using a Majorization–Minimization-based Bradley–Terry estimation. These merits are then mapped to a semantic-clarity score and a physical-affordance score. Their product is compared with a fixed decision-theoretic rough-set threshold derived from an explicit loss matrix. Boundary cases are then handled by a selector that reasons about multi-target ambiguity, occlusion, and stacking. Deployed on the SO-100 platform with a dual-view controller, 3WD-VLA improves success on executable tasks relative to the evaluated baselines and yields more interpretable collaborative behavior under ambiguity, clutter, and physical constraints.

Keywords: Vision-language-action models · Three-way decision · Bradley–Terry model · Decision-theoretic rough sets · Safe manipulation

1 Introduction

Vision-language-action (VLA) models and imitation-learning policies have substantially advanced end-to-end robotic manipulation by mapping images, robot states, and language instructions directly to control actions [1]. This trend has made language-conditioned grasping increasingly practical, but it also exposes a deployment gap: most policies are optimized for direct execution and implicitly

assume that the instruction is precise, the target is visible, the target is physically accessible for grasping, and the workspace is safe [9,27].

These assumptions often fail in shared human-robot workspaces. A command such as "grasp the blue cube" may be ambiguous when multiple identical cubes are present. Even with an unambiguous target, the object may be fully occluded or physically inaccessible because another object blocks the grasp path or is stacked on top of it. Human-hand intrusion into the robot workspace poses a safety-critical condition in shared human–robot environments [27]. Ambiguity, occlusion, and stacking may be resolved through clarification or recovery planning, whereas detection of human-hand intrusion must block subsequent grasp execution and trigger an emergency stop. Target absence is another negative-domain condition: when the requested object is absent, the system should abort the task rather than attempt to grasp without a valid target. Handling these conditions requires more than accurate perception: the system must choose among direct execution, user clarification, recovery planning, and safe rejection. A practical collaborative grasping system must therefore decide how to act, whether immediate execution is safe and appropriate, when to ask for clarification, and when to replan.

Existing approaches address parts of this problem, but they usually treat them separately. End-to-end VLA or imitation-learning policies can map visual observations and language instructions to robot actions, but they often assume that the requested task is immediately executable [1]. Safety-aware methods can prevent unsafe execution, but they mainly focus on stopping or rejecting risky actions rather than distinguishing recoverable boundary cases [27]. Clarification-based human–robot interaction methods can request user input when the language goal is ambiguous, but they do not provide a unified routing mechanism for semantic ambiguity, physical occlusion, stacking, target absence, and hand intrusion [6,20]. A unified mechanism for routing scenes to direct execution, user clarification, recovery planning, or safe rejection is therefore needed.

To address this gap, we proposed 3WD-VLA, a hierarchical dual-view framework that combines a vision-language supervisor, a mathematically grounded three-way decision core, and a boundary-case selector [22,24,26]. The key idea is to separate scenes into negative, boundary, and positive domains. Negative-domain cases require an emergency stop or task abortion for safety or validity reasons. Positive-domain samples are directly executable. Boundary cases are valid task instances with an unresolved prerequisite for direct execution, such as target disambiguation, obstacle clearing, or top-object removal. These cases are therefore deferred to clarification or recovery planning.

The detailed pipeline and sequential routing logic of 3WD-VLA are presented in Section 3. In brief, the framework first filters hard negative cases, then separates directly executable cases from recoverable boundary cases, and finally maps boundary cases to clarification or recovery actions before robot execution.

This work makes three main contributions. First, we adapt the classical three-way decision principle to collaborative robotic grasping and develop it into a 3WD-inspired hierarchical routing framework for VLA-based manipulation. Here, "hierarchical routing" means that the vision-language model is not used

only as a direct action predictor; instead, it acts as a high-level decision layer that routes each scene to safe rejection, direct execution, user clarification, or recovery planning before the low-level controller executes the selected action. The proposed framework separates unsafe or infeasible negative cases, confidently executable positive cases, and recoverable boundary cases that require clarification or replanning, thereby avoiding premature execution under uncertainty.

Second, we develop a decision core that combines the Bradley–Terry (BT) model with decision-theoretic rough sets (DTRS). It converts pairwise preferences generated by a fine-tuned vision-language reasoning module over four high-level actions (direct execution, asking for clarification, replanning, and rejection) into latent BT merits. A threshold derived from a DTRS-style loss matrix then enables cost-sensitive routing.

Third, we deploy the full system on an SO-100 robotic platform and demonstrate improved performance on executable tasks and more interpretable collaborative behavior compared with the evaluated controller baselines.

2 Related Work

2.1 VLA and Imitation Learning for Manipulation

Imitation learning has long provided a practical approach to language-conditioned manipulation by enabling robots to learn mappings from visual observations, robot states, and task descriptions to control actions using demonstrations. Early work incorporated free-form natural language into visuomotor imitation learning, while later studies improved long-horizon and multi-task performance through hierarchical representations, latent plans, and multimodal transformers. More recent manipulation policies such as ACT [29] and Diffusion Policy [4] further improved precision and robustness by modeling action chunks or multimodal action distributions. In the context of semantic grasping, KITE [17] and GraspCLIP [18] showed that language should influence not only which object is selected, but also how it should be grasped with respect to task semantics and affordances [11,16].

Recent work has extended this direction toward foundation models for robot control. PaLM-E [7] embeds continuous robot observations into a large embodied multimodal language model, while RT-1 [2] explored large-scale transformer policies for real-world control. RT-2 [30] explicitly unifies vision, language, and action to transfer web knowledge into robotic control. OpenVLA [10] provides an open-source VLA model that supports efficient fine-tuning. π_0 [1] introduces a flow-matching VLA architecture for general robot control. These models have substantially improved semantic grounding, cross-task generalization, and the ability to follow natural-language commands. However, they are largely optimized for direct action generation from the current observation and instruction, rather than for an explicit collaborative layer that decides whether the robot should execute immediately, ask for clarification, replan, or reject the task.

2.2 Safety- and Ambiguity-Aware Collaborative Manipulation

Safety-aware manipulation has been studied through control barrier functions, shielding, and reachability-based safeguards [14,19,20]. More recent work has augmented physical safety with semantic reasoning or failure detection, for example through semantic safety filters, failure-aware vision-language model (VLM) reasoning, and safety-aligned vision-language-action (VLA) models[3,8,9,27]. In parallel, ambiguity-aware human–robot interaction has explored clarification questions and interactive grounding for under-specified language [5,6,12,13,15,21,28]. These lines of work are complementary: the former emphasizes safe execution, while the latter focuses on clarification when the language goal is unclear.

2.3 Three-Way Decision

Three-way decision (3WD) addresses uncertainty through three actions: acceptance, deferment, and rejection. Decision-theoretic rough sets (DTRS) instantiate this idea by deriving decision thresholds from a loss matrix, rather than selecting thresholds heuristically [22,23,24,25,26]. This makes domain assignment sensitive to decision costs: a high loss can be assigned to erroneous direct execution on non-positive scenes, while a smaller loss can be assigned to unnecessary deferment on positive scenes.

In this work, we use this perspective as the decision-theoretic basis for routing collaborative grasping scenes. Unlike standard direct-execution policies, our framework separates hard negative cases, directly executable positive cases, and recoverable boundary cases that require clarification or recovery planning. Section 3 presents the sequential routing pipeline.

3 Method

3WD-VLA is a sequential three-way decision framework that combines a dual-view reasoning module with a robot-side controller. Figure. 1 summarizes the overall pipeline. Given a top-view image, a wrist-view image, and a language instruction, the system first merges the two visual observations into a dual-view representation. Instead of directly predicting a low-level robot command, the reasoning module generates structured decision text that describes both *what* the robot should do and *why*. The generated text is parsed into supervisor states, pairwise action preferences among `direct`, `ask`, `replan`, and `reject`, and boundary-case descriptors. These parsed outputs are processed in three sequential stages: a supervisor for hard negative cases, a Bradley–Terry+DTRS decision core for positive-versus-boundary routing, and a boundary selector for clarification or recovery actions. Finally, the selected high-level action code is passed to the robot-side controller for execution. The framework thus determines whether to execute directly, defer the task, or perform a recovery action before grasping.

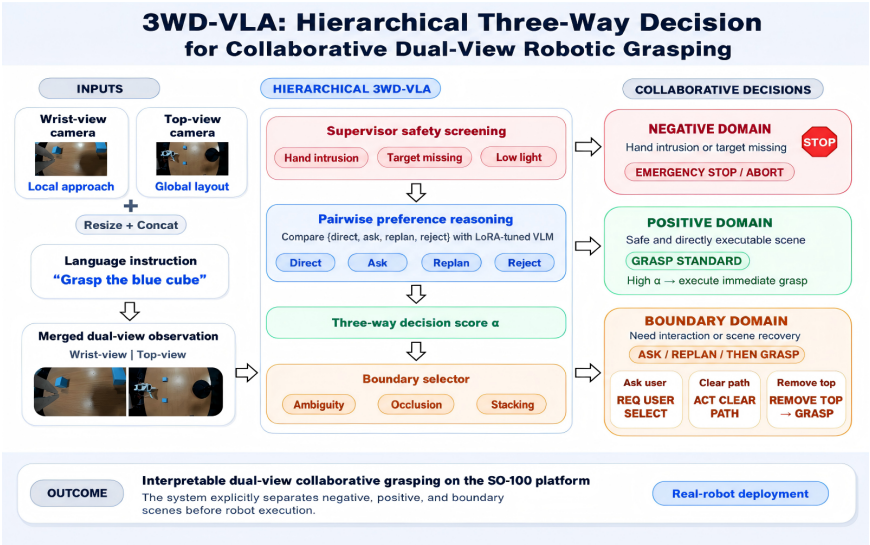


Fig. 1. Overview of 3WD-VLA.

3.1 Sequential Three-Way Routing Pipeline

At step t, the input consists of a top-view image I_t^{top}, a wrist-view image I_t^{wrist}, and a language instruction T. Instead of directly predicting a low-level robot command, the reasoning model first produces a structured response that describes the preferred decision and its rationale:

$$\hat{Y}_t = [\hat{C}_t \parallel \hat{R}_t], \tag{1}$$

where $\hat{C}_t$ denotes the pairwise comparison statements and $\hat{R}_t$ denotes the safety reasoning for the current scene. We compare direct execution with each of three alternative decisions: ask, replan, and reject. Thus, the candidate decision set is

$$\mathcal{H} = \{direct, ask, replan, reject\}. \tag{2}$$

In the three-way decision formulation, let $x_t = (I_t^{\text{top}}, I_t^{\text{wrist}}, T)$ denote the current dual-view scene and language instruction.

The routing layer maps x_t to an ordered list of high-level action codes L_t, whose length is denoted by K_t. For terminal or directly executable cases, $K_t = 1$. For recoverable boundary cases, $K_t > 1$ may be required because the robot may need to perform clarification or recovery actions before the final grasp.

The input space is partitioned into three domains: the negative domain $\mathcal{N}$, the positive domain $\mathcal{P}$, and the boundary domain $\mathcal{B}$. The negative domain contains hard-negative cases, such as target absence or safety-critical hand intrusion. The positive domain contains cases judged to be safe and directly executable with high confidence. The boundary domain contains recoverable cases that require

clarification or physical recovery. The routing policy is defined as

$$L_t = \Pi(x_t) = \begin{cases} [\Pi_{\text{neg}}(x_t)], & x_t \in \mathcal{N}, \\ [\text{GRASP_STANDARD}], & x_t \in \mathcal{P}, \\ \Pi_{\text{sel}}(x_t), & x_t \in \mathcal{B}. \end{cases} \tag{3}$$

Here, Π_{neg} is the negative-domain handler, which maps hard-negative cases to terminal actions such as `emergency-stop` for hand intrusion or `abort-target-missing` for target absence. `grasp-standard` denotes the standard direct grasping action. Π_{sel} is the boundary selector, which maps recoverable boundary cases to ordered action lists such as `req-user-select`, `act-clear-path`, `grasp-standard`, or `act-remove-top`, `grasp-bottom`.

A single camera view is often insufficient for collaborative grasping because the target object may be occluded or partially visible from one view. Therefore, we use a dual-view representation by merging the top-view and wrist-view images:

$$I_t^{merge} = \text{Concat}(\text{Resize}(I_t^{top}), \text{Resize}(I_t^{wrist})). \tag{4}$$

The reasoning module uses the merged dual-view image and the language instruction to generate the structured response.

The reasoning module is fine-tuned on a custom dataset of dual-view observations paired with instructions and structured responses. Each sample pairs dual-view observations with a dialogue-style target response composed of pairwise decision logic and safety reasoning. This supervision teaches the model not only to choose a suitable action, but also to justify its decision using visual and safety cues.

Algorithm 1 summarizes the deployed sequential routing logic.

Algorithm 1 Sequential 3WD-VLA inference pipeline

Input: dual-view images $I_t^{\text{top}}, I_t^{\text{wrist}}$, instruction T
Output: ordered action list L_t

$I_t^{\text{merge}} \leftarrow \textsc{MergeViews}(I_t^{\text{top}}, I_t^{\text{wrist}})$
$(z_t^{\text{sup}}, c_t^{\text{sup}}) \leftarrow \textsc{Supervisor}(I_t^{\text{merge}})$
if $z_t^{\text{sup}} = N_{\text{hand}}$ **then**
    return [`EMERGENCY_STOP`]
else if $z_t^{\text{sup}} = N_{\text{miss}}$ **then**
    return [`ABORT_TARGET_MISSING`]
end if
if $c_t^{\text{sup}} \neq \varnothing$ **then**
    $\textsc{ApplySafetyConstraint}(c_t^{\text{sup}})$
end if
$(z_t^{\text{3wd}}, \text{Score}_t, \pi_t) \leftarrow \textsc{ThreeWayCore}(I_t^{\text{merge}}, T, \alpha_{\text{DTRS}})$
if $z_t^{\text{3wd}} = \mathcal{P}$ **then**
    return [`GRASP_STANDARD`]
end if
$L_t \leftarrow \textsc{BoundarySelector}(I_t^{\text{merge}}, T)$
return L_t

3.2 Supervisor and Bradley–Terry+DTRS Decision Core

The supervisor filters out hard negative-domain cases before the BT+DTRS core performs further routing. It is implemented as a separate VLM query with a dedicated supervisor prompt applied to the merged dual-view observation. The query checks safety and task validity by predicting three Boolean indicators,

$$h_t, m_t, d_t \in \{0, 1\}, \tag{5}$$

which indicate human-hand intrusion, target absence, and severely degraded lighting, respectively. The supervisor state is

$$z_t^{\text{sup}} \in \{N_{\text{hand}}, N_{\text{miss}}, B_{\text{light}}, P_{\text{sup}}\}. \tag{6}$$

The state is assigned as

$$z_t^{\text{sup}} = \begin{cases} N_{\text{hand}}, & h_t = 1, \\ N_{\text{miss}}, & h_t = 0 \wedge m_t = 1, \\ B_{\text{light}}, & h_t = 0 \wedge m_t = 0 \wedge d_t = 1, \\ P_{\text{sup}}, & h_t = 0 \wedge m_t = 0 \wedge d_t = 0. \end{cases} \tag{7}$$

Here, N_{hand} triggers an emergency stop, N_{miss} triggers task abortion when the requested target is absent, B_{light} denotes a supervisor-level boundary state under degraded visibility, and P_{sup} indicates that the sample can proceed to the pairwise decision core. The supervisor query also returns an optional safety constraint c_t^{sup}; when this output is non-empty, the corresponding constraint is applied before downstream routing.

For the remaining samples, pairwise comparisons generated by the vision-language model (VLM) are encoded in a win-count matrix. The Bradley–Terry model then estimates latent action merits from these comparisons. Let

$$\mathcal{H} = \{\text{direct}, \text{ask}, \text{replan}, \text{reject}\} \tag{8}$$

be the set of candidate high-level actions. For any two distinct actions $i, j \in \mathcal{H}$, the notation $i \succ j$ means that action i is preferred over action j according to the parsed VLM comparison. The Bradley–Terry model associates each action i with a positive latent merit $\pi_i > 0$. Given the merit vector $\pi = (\pi_i)_{i \in \mathcal{H}}$, the probability that action i is preferred over action j is

$$\Pr_{\text{BT}}(i \succ j \mid \pi) = \frac{\pi_i}{\pi_i + \pi_j}. \tag{9}$$

The parsed pairwise comparisons are stored in a win-count matrix $W_t = [w_{ij,t}]$, where $w_{ij,t} \in \mathbb{N}_0$ is the number of parsed comparison statements at time t in which action i is preferred over action j. For a single structured response, $w_{ij,t}$ is typically either 0 or 1, while the notation also allows repeated or aggregated

comparisons. Let $\mathcal{E}_t = \{\{i,j\} : i,j \in \mathcal{H}, i \neq j, w_{ij,t} + w_{ji,t} > 0\}$ denote the set of action pairs with observed comparisons. The Bradley–Terry likelihood is

$$\mathcal{L}_{\mathrm{BT}}(\pi_t; W_t) = \prod_{\{i,j\} \in \mathcal{E}_t} \left(\frac{\pi_{i,t}}{\pi_{i,t} + \pi_{j,t}} \right)^{w_{ij,t}} \left(\frac{\pi_{j,t}}{\pi_{i,t} + \pi_{j,t}} \right)^{w_{ji,t}}. \tag{10}$$

and the log-likelihood is

$$\ell(\pi) = \sum_{i<j} \left[w_{ij} \log \pi_i + w_{ji} \log \pi_j - (w_{ij} + w_{ji}) \log(\pi_i + \pi_j) \right]. \tag{11}$$

The total win count for action i is defined as

$$\omega_i = \sum_{j \in \mathcal{H} \setminus \{i\}} w_{ij} \tag{12}$$

and the total number of comparisons between actions i and j is

$$n_{ij} = w_{ij} + w_{ji}. \tag{13}$$

Direct maximization is complicated because of the coupled $\log(\pi_i + \pi_j)$ terms. We therefore use a Majorization–Minimization (MM) fixed-point iteration. Since $-\log x$ is convex, for any $x > 0$ and $y > 0$,

$$-\log(x) \geq -\log(y) - \frac{x - y}{y}, \tag{14}$$

where equality holds when $x = y$. In the BT objective, we set $x = \pi_i + \pi_j$ and $y = \pi_i^{(s)} + \pi_j^{(s)}$, where s is the MM iteration index. This yields

$$\pi_i^{(s+1)} = \frac{\omega_i}{\sum_{j \in \mathcal{H} \setminus \{i\}} \frac{n_{ij}}{\pi_i^{(s)} + \pi_j^{(s)}}}. \tag{15}$$

After each iteration, the merits are normalized so that $\sum_{i \in \mathcal{H}} \pi_i = 1$. After convergence, the normalized BT merit vector for the current scene is

$$\boldsymbol{\pi}_t = [\pi_{\mathrm{direct},t}, \pi_{\mathrm{ask},t}, \pi_{\mathrm{replan},t}, \pi_{\mathrm{reject},t}],$$

where each component denotes the latent merit of the corresponding candidate action in the fixed action set $\mathcal{H} = \{\mathrm{direct}, \mathrm{ask}, \mathrm{replan}, \mathrm{reject}\}$. These merits are projected onto two interpretable axes,

$$P_{\mathrm{rel}} = \frac{\pi_{\texttt{direct}}}{\pi_{\texttt{direct}} + \pi_{\texttt{ask}}}, \qquad P_{\mathrm{aff}} = \frac{\pi_{\texttt{direct}}}{\pi_{\texttt{direct}} + \pi_{\texttt{replan}} + \pi_{\texttt{reject}}}, \tag{16}$$

and combined into

$$\mathrm{Score}_t = P_{\mathrm{rel}} \times P_{\mathrm{aff}}. \tag{17}$$

Intuitively, P_{rel} decreases when semantic ambiguity favors ask, while P_{aff} decreases when physical obstruction or stacking favors replan or reject.

Table 1. Loss design used to derive the fixed DTRS threshold.

Symbol	Value	Decision case	Role
λ_{PP}	0.05	positive scene, decide positive	preferred correct action
λ_{BP}	0.20	positive scene, decide boundary	unnecessary delay
λ_{PN}	0.90	non-positive scene, decide positive	unsafe direct execution
λ_{BN}	0.05	non-positive scene, decide boundary	safe deferment

We use DTRS to derive a fixed threshold for separating positive and boundary cases based on the score in Eq. (17). Let a_P denote direct execution and a_B denote deferment to clarification or replanning. If λ_{ij} is the loss of taking action $a_i \in \{a_P, a_B\}$ when the latent scene state is $j \in \{P, N\}$, where the first subscript denotes the selected action and the second subscript denotes the latent scene state, the DTRS threshold is

$$\alpha_{\mathrm{DTRS}} = \frac{\lambda_{PN} - \lambda_{BN}}{(\lambda_{PN} - \lambda_{BN}) + (\lambda_{BP} - \lambda_{PP})}. \tag{18}$$

The fixed loss in Table 1 yields $\alpha_{\mathrm{DTRS}} = 0.85$, so the routing rule is

$$z_t^{\mathrm{3wd}} = \begin{cases} \mathcal{P}, & \mathrm{Score}_t \geq \alpha_{\mathrm{DTRS}}, \\ \mathcal{B}, & \mathrm{Score}_t < \alpha_{\mathrm{DTRS}}. \end{cases} \tag{19}$$

Classical DTRS uses two thresholds (α, β), but our sequential formulation omits an explicit negative-domain routing decision from the second stage: once a sample reaches the Bradley–Terry+DTRS core, hard-negative cases have already been filtered by the supervisor.

Algorithm 2 summarizes the pairwise decision core.

3.3 Boundary Selector and Robot-Side Execution

Samples with $\mathrm{Score}_t < \alpha_{\mathrm{DTRS}}$ are treated as *recoverable* boundary cases. The selector extracts

$$r_t = (n_t^{top}, n_t^{wrist}, n_t, o_t, s_t), \tag{20}$$

where n_t^{top} and n_t^{wrist} are target counts from the top and wrist views, $n_t = \max(n_t^{top}, n_t^{wrist})$ is the consolidated count, $o_t \in \{0, 1\}$ indicates occlusion, and $s_t \in \{0, 1\}$ indicates stacking. The target counts are estimated directly by Qwen3-VL as part of the selector reasoning task; no separate object detector or rule-based counting module is used. The selector then uses these VLM-estimated counts together with the occlusion and stacking indicators to apply three rules,

$$\begin{aligned} n_t > 1 \wedge \neg q(T) &\Rightarrow \texttt{REQ_USER_SELECT}, \\ s_t = 1 &\Rightarrow [\texttt{ACT_REMOVE_TOP}, \texttt{GRASP_BOTTOM}], \\ o_t = 1 &\Rightarrow [\texttt{ACT_CLEAR_PATH}, \texttt{GRASP_STANDARD}]. \end{aligned} \tag{21}$$

If none of these conditions holds, the selector falls back to GRASP-STANDARD.

Algorithm 2 Three-way decision core based on pairwise preferences

Input: merged image I_t^{merge}, instruction T, threshold α_{DTRS}
Output: domain label z_t^{3wd}, score Score_t, merit vector π_t

$\mathcal{H} \leftarrow \{\texttt{direct}, \texttt{ask}, \texttt{replan}, \texttt{reject}\}$
$\hat{Y}_t \leftarrow \textsc{PairwiseReasoning}(I_t^{\text{merge}}, T)$
$\hat{C}_t \leftarrow \textsc{ParseComparisons}(\hat{Y}_t)$
$W \leftarrow \textsc{BuildWinMatrix}(\hat{C}_t, \mathcal{H})$
initialize $\pi^{(0)} \leftarrow \frac{1}{|\mathcal{H}|}\mathbf{1}$
for $s = 0$ to $S_{\max} - 1$ **do**
  for each $i \in \mathcal{H}$ **do**
    update $\pi_i^{(s+1)}$ by Eq. (15)
  end for
  normalize $\pi^{(s+1)}$ so that $\sum_i \pi_i^{(s+1)} = 1$
  if $\|\pi^{(s+1)} - \pi^{(s)}\|_1 < \varepsilon$ **then**
    break
  end if
end for
$\pi_t \leftarrow \pi^{(s+1)}$
compute P_{rel} and P_{aff}
$\text{Score}_t \leftarrow P_{\text{rel}} \times P_{\text{aff}}$
if $\text{Score}_t \geq \alpha_{\text{DTRS}}$ **then**
  $z_t^{\text{3wd}} \leftarrow \mathcal{P}$
else
  $z_t^{\text{3wd}} \leftarrow \mathcal{B}$
end if
return $(z_t^{\text{3wd}}, \text{Score}_t, \pi_t)$

The selected code is then passed to the low-level controller. Let $s_t \in \mathbb{R}^6$ be the robot state and $u_t \in \mathbb{R}^6$ the continuous action; the execution layer is modeled as

$$u_t = f_{\text{exec}}\big(s_t, I_t^{\text{top}}, I_t^{\text{wrist}}, p(\ell_t)\big), \tag{22}$$

where $p(\ell_t)$ is the prompt or condition associated with the selected high-level action. This separates structured collaborative reasoning from low-level motion generation while keeping the controller compatible with standard dual-view robot policies.

3.4 Model Adaptation and Robot-Side Execution

We use Qwen3-VL-8B-Instruct, a vision-language model for joint image-text understanding, as the reasoning backbone and fine-tune it with LLaMA-Factory using low-rank adaptation (LoRA) rather than updating all model parameters. LoRA adapters are inserted into the language-side projection layers {`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`}. The LoRA rank is set to 32 and the scaling factor is set to 64. We use the `qwen2_vl` template in LLaMA-Factory to jointly encode images and text, and set the maximum sequence length to 2048.

Algorithm 3 Boundary-domain selector

Input: merged image I_t^{merge}, instruction T
Output: ordered action list L_t

$(n_t^{top}, n_t^{wrist}, o_t, s_t) \leftarrow \textsc{SelectorReasoning}(I_t^{\text{merge}}, T)$
$n_t \leftarrow \max(n_t^{top}, n_t^{wrist})$
$L_t \leftarrow [\,]$
if $n_t > 1$ **and** $\neg q(T)$ **then**
    Append `REQ_USER_SELECT` to L_t
end if
if $s_t = 1$ **then**
    Append `ACT_REMOVE_TOP` to L_t
    Append `GRASP_BOTTOM` to L_t
    return L_t
else if $o_t = 1$ **then**
    Append `ACT_CLEAR_PATH` to L_t
    Append `GRASP_STANDARD` to L_t
    return L_t
end if
if $L_t = [\,]$ **then**
    Append `GRASP_STANDARD` to L_t
end if
return L_t

Training uses a per-device batch size of 1 and 16 gradient accumulation steps, yielding an effective batch size of 16. The learning rate is 1.0×10^{-4}, we use a cosine learning-rate schedule with a warmup ratio of 0.1, and the model is trained for 5 epochs using bf16 mixed precision. We reserve 10% of the training set for validation, perform evaluation every 100 steps, save a checkpoint every 100 steps, and log training status every 5 steps. The reasoning module is optimized using the standard token-level autoregressive cross-entropy loss for supervised fine-tuning (SFT), without an additional custom loss term

$$\mathcal{L}_{\text{SFT}} = -\sum_{(X,Y)} \sum_{m \in \Omega(Y)} \log p_\theta(y_m \mid X, y_{<m}), \tag{23}$$

where $\Omega(Y)$ denotes the supervised token positions of the target response.

At inference time, the generated structured decision text is parsed into pairwise comparisons, safety indicators, and a high-level action code ℓ_t. The pairwise comparisons are encoded in a win-count matrix, which is used to estimate BT merits via the updates in Eq. (15) and normalization, and finally into Score_t by Eq. (17). The DTRS rule determines whether the sample is directly executable or should be passed to the selector. The selected action is then translated into robot-side control.

Fig. 2 expands the overview in Fig. 1 by detailing the reasoning-layer workflow: supervisor filtering, BT-based merit estimation, DTRS thresholding, and boundary-case selection.

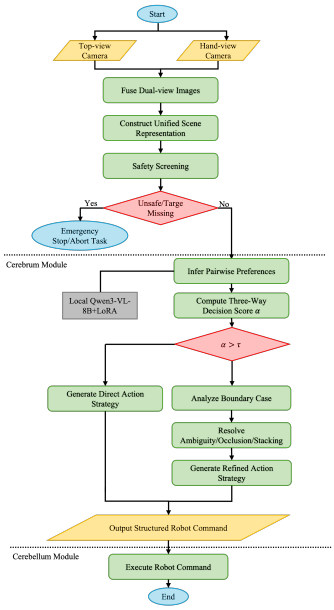


Fig. 2. Detailed decision flow used by the reasoning layer. The flowchart highlights the sequential supervisor filter, BT-based merit recovery, DTRS thresholding, and boundary-case selector.

4 Experimental Setup

4.1 Platform, Task Paradigms, Datasets, and Baselines

All controller-side experiments are conducted on the SO-100 robotic manipulation platform. Training and deployment validation are performed on a local workstation running Ubuntu 22.04.5 LTS, Linux kernel 6.8.0-94-generic, x86_64 CPU architecture, an NVIDIA GeForce RTX 4060 16GB GPU, NVIDIA Driver 575.64.03, and CUDA 12.9. All controllers use the same observation and action interface: a 6-dimensional robot state, two RGB images from the wrist and top cameras (raw size $3 \times 360 \times 640$), and a 6-dimensional action output. We use `n_obs_steps=1`, seed 1000, four data-loader workers, CUDA execution, and 50 rollout episodes per evaluation pass with `use_async_envs=false`.

We evaluate two task paradigms. In the *task-decomposition* setting, a high-level module decomposes the long-horizon task into subtasks and the controller executes the current subtask; 3WD-VLA and SmolVLA w/o 3WD (SmolVLA without 3WD) operate in this regime. In the *end-to-end long-horizon* setting, the controller uses the current observation to generate actions for the full long-horizon task; SmolVLA_xr, ACT, and π_0 are evaluated in this regime. The reasoning module is trained with `grasp_final_with_reasoning.jsonl` (5,998 samples), while controller datasets are collected on the real SO-100 platform. Task definitions are summarized in Table 2.

Our experimental setup includes four learned controllers: SmolVLA, SmolVLA_xr, ACT, and π_0. Table 3 summarizes the key training configurations of these con-

Table 2. Controller-side task configuration in the two evaluation paradigms.

Paradigm	Task 1	Task 2	Task 3
Decomposed subtask (3WD-VLA, SmolVLA w/o 3WD)	left / right / ambiguous blue-cube grasp 50 + 50 + 100 episodes	move the obstacle 100 episodes	pick up the top cube 100 episodes
End-to-end long horizon (SmolVLA_xr, ACT, π_0)	left / right blue-cube grasp 50 + 50 episodes	move the obstacle and grasp the blue cube 80 episodes	pick up the top cube and grasp the blue cube 80 episodes

Table 3. Key controller training configurations. All policies use dual-view RGB observations, a 6-D robot state, and a 6-D action output.

Method	Steps	LR	Chunk size	Vision frozen
SmolVLA	40k	1.0×10^{-4}	50	Yes
SmolVLA_xr	30k	1.0×10^{-4}	50	Yes
ACT	20k	1.0×10^{-5}	100	No
π_0	40k	2.5×10^{-5}	50	No

trollers.SmolVLA serves both as the low-level execution controller inside our full system and as the main ablation baseline when the upper three-way decision layer is removed. SmolVLA and SmolVLA_xr share the same core training interface (dual-view RGB input, 6-D state/action, chunk size 50, action horizon 50, frozen vision encoder, expert-only training, AdamW with cosine decay). ACT uses a Transformer action-chunking policy with a ResNet-18 visual backbone, model dimension 512, eight attention heads, feedforward dimension 3200, and a VAE latent action model with latent dimension 32 and KL weight 10. π_0 is fine-tuned from `lerobot/pi0_base` with bf16, gradient checkpointing, image resolution 224×224, and compiler optimization. The reasoning model is Qwen3-VL-8B-Instruct fine-tuned with LoRA under LLaMA-Factory; LoRA is injected into the language-side projection layers with rank 32 and scaling factor 64, using the `qwen2_vl` template and maximum sequence length 2048. The controller-side configuration details are drawn from the released training files, which keep the dual-view image interface, 6-D state/action shape, and 50-episode evaluation budget consistent across the compared policies. Concretely, SmolVLA and SmolVLA_xr resize images with padding to 512×512 and train only the expert module together with the state projection layer; ACT uses a ResNet-18 backbone initialized from ImageNet and a Transformer with one decoder layer with VAE action modeling; and π_0 uses bf16 training, gradient checkpointing, 10 denoising steps, and compile-mode acceleration. These configurations provide context for interpreting performance differences and controller design trade-offs.

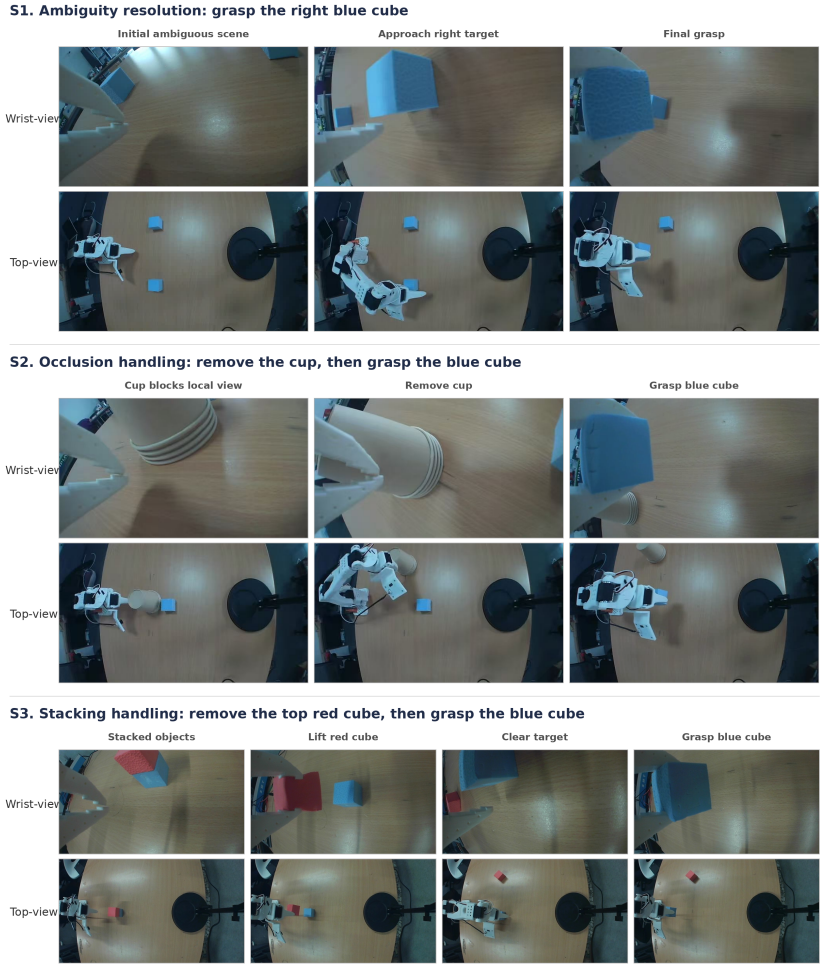


Fig. 3. Representative real-robot rollouts for the three boundary-scene families. Each block shows wrist-view observations in the first row and top-view observations in the second row. From top to bottom, the blocks show user disambiguation in a scene with two blue cubes, obstacle removal before grasping, and top-object removal before grasping the target below.

4.2 Metrics

We evaluate task performance using the success rate (SR). Let N_{all} denote the total number of evaluated rollout trials for a given task, scene, or aggregated setting, and let N_{success} denote the number of trials in which the robot successfully completes the required task objective. The success rate is defined as

$$\text{SR} = \frac{N_{\text{success}}}{N_{\text{all}}}, \tag{24}$$

Table 4. Successful trials on executable subtasks. Micro aggregates all 40 trials. Macro is the unweighted mean success rate (%) across the ambiguity, occlusion, and stacking families; ambiguity combines the left- and right-target trials.

Method	Amb.-L	Amb.-R	Occl.	Stack.	Micro	Macro
3WD-VLA (ours)	**8/10**	**9/10**	**8/10**	**7/10**	**32/40**	**78.3**
SmolVLA w/o 3WD	7/10	6/10	6/10	2/10	21/40	48.3
ACT	0/10	4/10	7/10	2/10	13/40	36.7
π_0	6/10	5/10	6/10	4/10	21/40	51.7

We define decision correctness (DC) as the proportion of trials with the expected high-level decision. Let $N_{\text{correct_decision}}$ denote the number of trials in which the high-level decision matches the expected decision label, and let N_{all} denote the total number of evaluated trials. Decision correctness is defined as

$$\text{DC} = \frac{N_{\text{correct_decision}}}{N_{\text{all}}}. \tag{25}$$

We additionally use ambiguity handling accuracy (AHA), recovery success rate (RSR), and safety stop or abort accuracy (SSA) when negative-domain episodes are available. Because the end-to-end baselines do not share exactly the same task semantics as the decomposed setting, the main quantitative tables focus on executable subtasks that are most directly comparable across methods.

5 Results

The quantitative results are based on real-world rollout records, with successful trials counted in blocks of 10. Individual entries therefore use a denominator of 10, while aggregated entries use denominators of 20 or 40. SmolVLA_xr is included in the training setup for completeness but is excluded from the quantitative tables because its real-world rollout evaluation remains incomplete.

5.1 Overall Success

We summarize success on executable subtasks to distinguish task execution from the complete collaborative decision process. These subtasks are left- or right-target grasping after disambiguation, obstacle removal followed by grasping in occluded scenes, and top-object removal followed by grasping in stacked scenes. Table 4 reports these results together with micro and macro summaries.

Overall, 3WD-VLA achieves the best recorded performance among methods with completed rollout logs, with a micro success rate of 80.0% (32/40) and a macro success rate of 78.3% . The most controlled comparison is the ablation against SmolVLA w/o 3WD, because both systems share the same decomposed-controller setting and differ mainly in whether the upper three-way decision layer is present.

Table 5. Scene/prompt-level success rate (%, with successful trials in parentheses) across the full 3WD-VLA system and non-3WD baselines. S1: two identical blue cubes; S2: blue cube fully occluded by a paper cup; S3: red/yellow cube stacked on top of the blue cube. A: ambiguous prompt; L/R: left/right prompt; M: move obstacle and grasp; T: pick up top cube and grasp.

Scene	Prompt	3WD-VLA(ours)	SmolVLA w/o 3WD	ACT	π_0
S1	A	60 (6/10)	0 (0/10)	50 (5/10)[†]	40 (4/10)
S1	L	80 (8/10)	70 (7/10)	0 (0/10)	60 (6/10)
S1	R	90 (9/10)	60 (6/10)	40 (4/10)	50 (5/10)
S2	A	50 (5/10)	30 (3/10)	70 (7/10)	40 (4/10)
S2	M	80 (8/10)	60 (6/10)	70 (7/10)	60 (6/10)
S3	A	60 (6/10)	50 (5/10)	20 (2/10)	30 (3/10)
S3	T	70 (7/10)	20 (2/10)	20 (2/10)	40 (4/10)

[†] In S1-A, ACT succeeds mainly on the right target and exhibits a clear right-side bias rather than genuine ambiguity resolution.

5.2 Boundary-Case Performance

The proposed hierarchy is designed to address recoverable boundary cases. Representative real-robot rollouts for ambiguity resolution, occlusion recovery, and stacking recovery are shown in Fig. 3. In the ambiguity scenes, the scene contains two visually identical blue cubes. On the executable subtasks after clarification, 3WD-VLA achieves 8/10 and 9/10 success on the left and right targets, outperforming SmolVLA w/o 3WD (7/10, 6/10), ACT (0/10, 4/10), and π_0 (6/10, 5/10). Under the original ambiguous prompt "grasp the blue cube," the non-3WD baselines do not exhibit genuine disambiguation behavior: SmolVLA w/o 3WD records 0/10, π_0 records 4/10, and ACT records 5/10, but the ACT rollout notes reveal a strong right-side bias rather than true clarification behavior.

For occlusion, 3WD-VLA succeeds in 8 of 10 trials, compared with 7/10 for ACT and 6/10 for both SmolVLA w/o 3WD and π_0. For stacking, the improvement is larger: 3WD-VLA succeeds in 7 of 10 trials, compared with 4/10 for π_0 and 2/10 for both ACT and SmolVLA w/o 3WD. This is consistent with the explicit boundary recovery plan `ACT_REMOVE_TOP`→`GRASP_BOTTOM`, a sequence that direct policies struggle to execute reliably in the evaluated scenes.

Table 5 further shows that the strongest difference between 3WD-VLA and direct baselines appears in prompts requiring collaboration and scenes with physical constraints. The gap is particularly clear when the same visual scene can be interpreted as either a clarification case or a one-shot execution case depending on the language instruction.

5.3 Ablation Studies

The most controlled comparison in the paper isolates the contribution of the three-way routing layer by comparing the full 3WD-VLA system against the same SmolVLA controller without upper-layer routing. Table 6 summarizes the result.

Table 6. Ablation of the 3WD hierarchy relative to the same SmolVLA controller without upper-layer routing. Ambiguity aggregates the left/right executable subtasks.

Setting	Ambiguity	Occlusion	Stacking	Micro
3WD-VLA (ours)	**17/20**	**8/10**	**7/10**	**32/40**
SmolVLA w/o 3WD	13/20	6/10	2/10	21/40
Gain (pts.)	**+20.0**	**+20.0**	**+50.0**	**+27.5**

The full model improves the SmolVLA backbone from 21/40 to 32/40 on the micro summary, with the largest gain on stacking. This indicates that the main benefit of 3WD-VLA is not merely refusing unsafe execution, but decomposing physically constrained scenes into recoverable intermediate actions before the final grasp.

5.4 Qualitative Analysis

Direct policies tend to collapse semantic ambiguity, physical inaccessibility, and execution risk into a single immediate grasp attempt. In the ambiguity scene, this appears as persistent side bias or missed target selection. In the occlusion scene, it appears as accidental obstacle collisions, incomplete clearing, or failure to re-localize the cube after the obstacle has moved. In the stacking scene, it appears as grasping both objects together, using excessive grasp depth, or producing unstable off-center grasps. Table 7 summarizes these typical failure modes and their representative symptoms.

3WD-VLA explicitly distinguishes these conditions. The top view is more useful for counting candidate targets and identifying stacking relationships, while the wrist view is more informative for assessing the local gripper approach and detecting obstructions along the direct path. More importantly, the reasoning model produces human-readable pairwise decision logic and safety reasoning before controller execution. The system first assigns the scene to a negative, boundary, or positive regime and then emits a structured action code such as `REQ_USER_SELECT`, `ACT_CLEAR_PATH`, or `ACT_REMOVE_TOP`. Such outputs are easier to audit and more aligned with collaborative manipulation in shared workspaces.

Table 7. Typical failure modes.

Scene	Method	Modes	Representative symptoms
S1	SmolVLA w/o 3WD	F1, F2	Unable to localize the intended blue cube consistently; off-center grasps frequently clip the edge and eject the object.
S1	ACT	F1	Fails to recognize the left target in the scene with two blue cubes and shows a strong right-side bias.
S2	SmolVLA w/o 3WD	F3, F6, F1	Occasionally displaces the obstacle through incidental contact, but still fails to re-localize the blue cube; some rollouts stall above the obstacle without completing the recovery.
S2	ACT	F3, F1, F2	Can partially move the obstacle, but the displacement is often insufficient; once the cube shifts, re-localization and stable grasping both become difficult.
S3	SmolVLA w/o 3WD	F4, F5, F2	May lift the upper and lower cubes together; after clearing the top object, off-center grasps on the blue cube often cause it to spring away.
S3	ACT	F4	Has difficulty reasoning about depth and stacked-object relations, and often grasps both objects simultaneously.

F1: target localization error; F2: unstable grasp contact; F3: insufficient obstacle removal; F4: depth/stacking misinterpretation; F5: multi-object coupling; F6: action stall.

6 Discussion and Limitations

The results suggest that the main value of 3WD-VLA is not simply a higher success count, but a more appropriate decision structure for collaborative manipulation. In the evaluated scenes, semantic ambiguity, partial visibility, object interference, and execution risk often appear together. A direct policy must generate control actions despite these uncertainties, whereas 3WD-VLA explicitly separates safety filtering, positive-versus-boundary routing, and boundary recovery.

A second strength is that the routing decision is mathematically interpretable. The VLM produces pairwise judgments, which the BT model converts into latent merits. The routing threshold is then derived from a DTRS loss matrix rather than selected directly as an ad hoc constant. This formulation nevertheless relies on assumptions: the pairwise comparisons must be sufficiently self-consistent, and the fixed loss design may not transfer unchanged to other applications.

Structured textual supervision offers both benefits and limitations. It makes the decision process more inspectable than a single end-to-end policy score, but it also means that downstream execution depends on stable parsing of generated text. Prompt sensitivity, output-format drift, or partially inconsistent reasoning traces can therefore affect controller stability.

The dual-view design also contributes to interpretability. The top view provides global evidence such as object count, approximate target layout, and stacking relationships, while the wrist-view image is more informative for local approach geometry and direct-path accessibility. This division is well aligned with the selector logic in our method.

Several limitations remain. First, overall performance remains limited by the robustness of low-level execution: localization errors, unstable contacts, and errors accumulated during recovery can still cause failure even when the high-level decision is correct. Second, the benchmark is not perfectly symmetric across methods, because 3WD-VLA and SmolVLA w/o 3WD are evaluated in a decomposed-subtask regime, whereas ACT and π_0 are evaluated end-to-end. Third, the study does not include a dedicated quantitative evaluation of negative-domain cases. Future evaluations should include hand-intrusion and target-absence episodes and report metrics such as safety-stop accuracy and the false-stop rate.

The study also lacks a systematic sensitivity analysis of the BT+DTRS layer. In this paper we fix $(\lambda_{PP}, \lambda_{BP}, \lambda_{PN}, \lambda_{BN}) = (0.05, 0.20, 0.90, 0.05)$, which yields $\alpha_{\mathrm{DTRS}} = 0.85$ and encodes a conservative preference against unsafe direct execution. Future work should examine loss-matrix sensitivity, MM convergence diagnostics, and calibration of Score_t under distribution shift, while also extending the benchmark to richer human-in-the-loop and safety-critical settings.

A further limitation lies in scale and diversity. The controller datasets are collected on a single platform, with a narrow object vocabulary centered on blue-cube tasks and a small set of structured obstacle patterns. This is sufficient for a proof-of-concept study, but it is not enough to claim strong generalization across object categories, materials, lighting conditions, or richer clutter distributions. Likewise, the current selector deliberately focuses on three boundary families: ambiguity, occlusion, and stacking. Extending it to more diverse boundary cases remains future work.

The ambiguity evaluation does not yet assess the complete collaborative interaction. The present experiments show whether the system can succeed on executable subtasks after clarification, but they do not yet comprehensively evaluate the interaction loop: whether the robot asks at the correct moment, whether the clarification request is understandable to the user, how quickly the user responds, and whether the post-clarification execution succeeds. Evaluating this complete interaction loop is an important direction for extending the benchmark.

7 Conclusion

We presented 3WD-VLA, a hierarchical dual-view framework that formulates collaborative robotic grasping as a sequential three-way decision problem. The framework integrates supervisor-based negative filtering, Bradley–Terry merit recovery from pairwise VLM judgments, a DTRS threshold for positive-boundary routing, and selector-based recovery planning for ambiguity, occlusion, and stacking.

Deployed on the SO-100 platform, 3WD-VLA improves performance on executable tasks relative to the evaluated baselines and yields more interpretable collaborative behavior. The results support the value of explicitly distinguishing negative, boundary, and positive domains, while also highlighting the need for larger-scale safety evaluation, stronger low-level execution robustness, and more systematic calibration of the BT+DTRS decision layer.

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